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### Flow controlled sound generation apparatus

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#### Abstract

A flow controlled sound generation system is disclosed that includes one or more fluid pumps to control air flow through a sound channel. The air flow is modulated through one or more valves to produce audible frequency pressure waves.

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## Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/687,479, filed Mar. 4, 2022, which is a continuation of U.S. patent application Ser. No. 16/888,331, filed May 29, 2020, now U.S. Pat. No. 10,679,597, which is a continuation of U.S. patent Ser. No. 15/812,050, filed Nov. 14, 2017, now U.S. Pat. No. 10,679,597, which is a continuation of U.S. application Ser. No. 14/145,783, filed Dec. 31, 2013, now U.S. Pat. No. 9,818,391, which claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 61/748,050 entitled “Flow Controlled Sound Generator,” filed Dec. 31, 2012; all of the aforementioned priority applications being hereby incorporated by reference in their respective entirety.

## BACKGROUND

(1) Low frequency sound is typically generated using moving diaphragms such as speakers or high voltage electrostatic diaphragms. Audio systems normally utilize a large diaphragm with a sealed frame to generate low frequency pulses.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosure herein is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings and in which like reference numerals refer to similar elements, and in which:
- (2) FIG. 1 is an example block diagram of a flow controlled sound generation system;
  - (3) FIG. 2 depicts an example apparatus of a flow controlled sound generation system;
  - (4) FIG. 3 is a flow chart depicting a method of generating sound by modulating air flow;
  - (5) FIG. 4A illustrates an example of a single array of air gates for a fluid pump;
  - (6) FIG. 4B illustrates an example of an opposing gate array for a bidirectional fluid pump;
  - (7) FIGS. 5A and 5B depict examples of sheet arrays included in a fluid pump to produce an air flow;
  - (8) FIGS. 6A and 6B illustrate a bidirectional arrangement for a sheet array included in a fluid pump to produce a bidirectional air flow;
  - (9) FIG. 7A illustrates a wave grid showing a fan-like structure which is rotationally actuated to cause flow;
  - (10) FIG. 7B is a three dimensional view of FIG. 7A with blades rendered for clarity of illustration;
  - (11) FIGS. 8A and 8B illustrate chamber-based modulators for producing phase-controlled air flow;
  - (12) FIG. 9 illustrates an implementation of a chamber-based modulator for producing a phase-controlled air flow; and
  - (13) FIG. 10 is an electrical schematic for a four-chambered modulator.
  - (14) FIG. 11 illustrates an example of a pump apparatus mounted inside a housing.

### DETAILED DESCRIPTION

- (15) A sound generation system is disclosed that includes a controller and a fluid pump to produce an air flow through a sound channel. The sound generation system further includes a sound control medium disposed within the sound channel to modulate the air flow to produce pressure waves at a plurality of audible frequencies. Additionally or as an alternative, the fluid pump can modulate the air flow to produce a variety of pressure waves of different phases, frequencies, and/or amplitudes.
- (16) In one aspect, the system can be realized as a pump with a post modulator or it can be realized as a dynamic pump which uses high frequency control to modulate its output.
- (17) The sound control medium can be an array of micro-valves configured to be operated in unison to produce pressure waves at audible frequencies. For example, low frequency sound waves may be produced at up to 300 Hz or higher. Each micro-valve can be comprised of a single gate, or may include multiple chambers for refined flow modulation. Furthermore, controlled fluid pumps for gas or liquids can provide a high degree of pressure control to the point where audio information can be emitted into a pump fluid medium for use in audio playback applications, actuator applications, or propulsion. Thus, a system is provided to generate controlled modulated acoustic flows for generation of low frequency sound. Also, a modulating pump is provided for fluid flow applications.
- (18) One or more embodiments described herein provide that methods, techniques, and actions performed by a computing device are performed programmatically, or as a computer-implemented method. Programmatically, as used herein, means through the use of code or computer-executable instructions. These instructions can be stored in one or more memory resources of the computing

device. A programmatically performed step may or may not be automatic.

(19) One or more embodiments described herein can be implemented using programmatic modules or components of a system. A programmatic module or component can include a program, a sub-routine, a portion of a program, or a software component or a hardware component capable of performing one or more stated tasks or functions. As used herein, a module or component can exist on a hardware component independently of other modules or components. Alternatively, a module or component can be a shared element or process of other modules, programs or machines.

(20) Some embodiments described herein can generally require the use of computing devices, including processing and memory resources. For example, one or more embodiments described herein can be implemented, in whole or in part, using computing devices such as desktop computers, cellular or smart phones, personal digital assistants (PDAs), laptop computers, and tablet devices. Memory, processing, and network resources may all be used in connection with the establishment, use, or performance of any embodiment described herein (including with the performance of any method or with the implementation of any system).

(21) Furthermore, one or more embodiments described herein may be implemented through the use of instructions that are executable by one or more processors. These instructions may be carried on a computer-readable medium. Machines shown or described with figures below provide examples of processing resources and computer-readable mediums on which instructions for implementing embodiments can be carried and/or executed. In particular, the numerous machines shown with embodiments include processor(s) and various forms of memory for holding data and instructions. Examples of computer-readable mediums include permanent memory storage devices, such as hard drives on personal computers or servers. Other examples of computer storage mediums include portable storage units, such as CD or DVD units, flash memory (such as carried on smart phones, multifunctional devices or tablets), and magnetic memory. Computers, terminals, network enabled devices (e.g., mobile devices, such as cell phones) are all examples of machines and devices that utilize processors, memory, and instructions stored on computer-readable mediums. Additionally, embodiments may be implemented in the form of computer-programs, or a computer usable carrier medium capable of carrying such a program.

(22) FIG. 1 is an example block diagram of a flow controlled sound generation system **100**. The sound generation system **100** can be utilized in conjunction with an audio unit **160** and/or a traditional speaker system **170**. In variations, an audio signal **162** is generated by the audio unit **160** and outputted to the sound generation system **100**. A controller **110** processes the audio signal **162** and implements commands to a fluid pump **120** and/or a sound control medium **140**.

(23) The fluid pump **120** can be configured to generate a continuous and direct air flow **122** through a sound channel for modulation by the sound control medium **140**. In such variations, the fluid pump **120** can be configured to either pull air into the sound channel, or push air into the sound channel.

(24) Alternatively the fluid pump **120** can be configured to generate an alternating flow **124**. For example, the fluid pump **120** can be a slow modulated alternating current (AC) pump such as a circular pump. In such variations, the downstream sound control medium (e.g., a valve array), can be timed to be in phase with the AC pump. Furthermore, for AC implementations, pressurized chambers may be included to bridge periods of low flow times between pump cycles. The AC pump may be a “push-pull” pump, which can redirect airflow as input into the sound channel, or output through the fluid pump **120**. Such push-pull pumps can be controlled dynamically according to pump control signals **112** from the controller **110**.

(25) Still further, the fluid pump **120** can be any variety or combination of different pumps to generate dynamic flows or pulses as described in further detail below.

(26) The fluid pump **120** can be configured to generate either unidirectional or bidirectional airflows, and can further be configured to generate an output flow that can either be modulated or un-modulated. Furthermore, the fluid pump **120** can be ultrasonic and can further be capable of

modulating its power in real-time and/or utilize a post modulator, such as a gated apparatus, to achieve a higher degree of pressure control efficacy.

(27) In variations, a second fluid pump operating anti-parallel to the first air pump can be utilized. In such variations, the outputs of both pumps can be merged allowing for true positive and negative pressure amplitudes, and can be achieved through the use of one or more internal valves with varying tradeoff in distortion and/or loading. Certain variations may include a sound channel shaped to allow for flow convergence, and may further include one or more valve arrays to further refine generated pressure pulses.

(28) Furthermore, the fluid pump **120** can include multiple flow channels, where each flow channel can be optimized for an audio frequency band. For example, when a desired sound has a frequency of 125 Hz, air flow may be redirected to a particular flow channel that has been optimized for a frequency range between say 100-150 Hz. In such arrangements, other flow channels can be blocked to allow a single optimized flow for that particular frequency range.

(29) The sound control medium **140** can be disposed within the sound channel and can be coupled to receive the air flow from the fluid pump **120**. Furthermore, the sound control medium **140** can be controlled by way of valve control signals **114** from the controller **110**. The sound control medium **140** can be a single valve constrictor that “pinches” the airflow according to a desired sound amplitude and/or frequency. However, given that such a constrictor valve can cause distortion and back flow shock to the fluid pump **120**, an air regulator may be included to prevent damage to the fluid pump **120**. Alternatively, the sound control medium **140** may comprise a single valve redirector that redirects the air flow through one or more waste channels, thereby preventing any back flow that could otherwise damage the fluid pump **120**.

(30) In variations, the sound control medium **140** can be an array of valves, or micro-valves (e.g., sound channel array **142**) that can be operated in unison to produce pressure pulses to be outputted through the sound channel. In such variations, the sound control medium **140** can be a grating disposed within the sound channel. The grating can have a micro-valve array disposed on a plurality of openings of the grating, where the micro-valves of the micro-valve array can be configured to operate in unison to collectively manipulate the air flow to produce pressure waves at a plurality of audible frequencies.

(31) The sound control medium **140** can also include a waste channel coupled to the sound channel. The waste channel can output a waste flow caused by a collective constriction of the air flow by the micro-valve array. Furthermore, a waste channel grating can be included and disposed within the waste channel, the waste channel grating can also have a micro-valve array (i.e., waste channel array **144**) disposed on a plurality of openings of the waste channel grating. Thus, the controller **110** can generate valve control signals that dynamically and synchronously control both the sound channel array **142** and the waste channel array **144** in order to produce the desired sounds.

(32) FIG. 2 depicts an example apparatus of a flow controlled sound generation system. In describing elements of FIG. 2, reference may be made elements described with respect to FIG. 1. Referring to FIG. 2, a sound generation system **200** includes a fluid pump **202** to generate an air flow **210** through a sound channel **204**. A micro-valve array **206** may be included within the sound channel **204** to modulate the air flow **210** to produce audio pulses **212** of a desired frequency.

(33) A waste channel **208** can be coupled to the sound channel **204** to prevent back flow from damaging the fluid pump **202**. Furthermore, as discussed above, a second micro-valve array **214** can be included in the waste channel. Both micro-valve arrays **206**, **214** can be synchronized for optimal deliver of sound **212**.

(34) Any number of waste channels can be included, as well as any number of optimized sound channels. For example, certain sound/waste channel arrangements may be optimized for a certain frequency range. Such arrangements may include a single sound channel with a plurality of waste channels coupled thereto. Alternatively, multiple sound channels with any number of waste channels may be arranged and tuned to produce optimum audible sounds in a specified frequency

range.

(35) The micro-valve arrays **206**, **214** can also include any number of valves. As frequency increases, the valves are required to operate more rapidly. Smaller and smaller valves are able to operate more rapidly than larger valves. Thus, gratings may be used to dispose hundreds, thousands, even tens or hundreds of thousands of micro-electro-mechanical (MEM) valves to control the relatively large air flow **210** and produce the sound pulses **212**. In order to operate in unison, these MEM valves may be controlled via varying electric fields, magnetic fields, or otherwise.

(36) Furthermore, any number of waste channels **208** and sound channels **204** may be included to produce a wide swath of audio frequencies, from 0 Hz-300 Hz and beyond. For example, a series of sound channels can be coupled to the pump **202** and each can include a micro-valve array, and can be tuned to a certain frequency band. Furthermore, the micro-valves themselves may be tuned or adjusted in size to produce a desired sound quality at a desired frequency band.

(37) As an addition or alternative, the pump **202** itself may include a flow regulator to restrict and admit flow through each of the series of sound channels. Further still, multiple valve array and multiple pump combinations are contemplated that may be tuned to specified frequency bands. Accordingly, a complex system of air flow generation and various degrees of modulation and refinement is contemplated to produce sharp pulses of low frequency pressure waves.

(38) Further still, for AC implementations, some valve types (e.g., piezo) have a limited frequency direct flow response. In other words, such valve types can have difficulty in maintaining constriction. Accordingly, for piezo valves may be coupled in series so that in the aggregate, the correct flow constriction is produced through their combined path. Valve constriction may be intricately controlled to produce audible frequencies of a high quality by means of a precisely timed controller capable of synchronizing the low frequency sound generation apparatus with, for example, an audio unit such as a media player.

(39) FIG. **3** is a flow chart depicting a method of generating sound by modulating air flow. Referring the FIG. **3**, the method includes producing air flow in a sound channel (**310**). As discussed above, the air flow may be generated by any number of fluid pumps (e.g., push, pull, modulated, un-modulated, unidirectional, bidirectional, etc.). The air flow can be a continuous direct flow (**312**), or an alternating flow (**314**), and can further be pre-modulated or pulsed to allow for refined post-modulation.

(40) One or more micro-valve arrays may be actuated to produce pressure waves (**320**) at audible frequencies. For direct flow implementations, the one or more arrays can be configured to simply actuate in order to produce the pulse. However, for slow modulating alternating flows, the one or more arrays may be timed to be in sync according to the alternating flow. Furthermore, all micro-valves can be synchronized to operate in unison (**322**) in order to produce a desired frequency of pressure wave. Alternatively, certain sets of micro-valves in the array can be regulated, such as closed or shut down during periods when certain frequency bands require (**324**). For example, a frequency band of 150 Hz-200 Hz may be optimal when a portion of the array is not used to produce a higher quality sound.

(41) As audio signals are transmitted to the controller, the controller can dynamically transmit valve control signals to the one or more arrays in order to modulate the pressure pulses to change pitch (**330**). Modulation of the pressure waves may also be achieved via the fluid pump. For example, dynamically controllable pumps, such as a non-resonant rigid array pump with an oscillatory drive mechanism, can produce air flow almost instantaneously. Thus, such pumps may be utilized to work in conjunction with the one or more micro-valve arrays in order to produce a higher quality sound and/or more efficient sound production.

(42) FIG. **4A** illustrates an example of a single array of air gates **400** for a fluid pump. Such an array **400** can be driven by a high frequency source which can generate air flow that can be both amplitude and/or frequency modulated to produce pressure pulses. The high frequency source, such

as an oscillating driver, can be coupled to an anchored end **402** of the array **400**, and the fluid pump can generate a steady air flow by driving the anchored edge **402** of the array **400**. As the array **400** is driven, a floating end **404** can produce a steady air flow. By way of both power adjustment and/or chambered valves, the air flow can be modulated to produce pressure waves of varying frequency and amplitude (e.g., at a far lower frequency than the frequency of the high frequency source).

(43) FIG. **4B** illustrates an opposing gate array **410** for a bidirectional pump. The opposing gate array **410** can include a source array **420** and a sink array **430**. Each array can be driven by a high frequency source to produce air flow in either direction or both directions simultaneously.

Accordingly, the driven source array **420** may be a source pump, and the driven sink array **430** may be a sink pump. Thus, when a source flow is required, the sink pump can be stopped. Alternatively, when a sink flow is required, the source pump can be stopped. Furthermore, both source and sink pumps may be driven to produce bidirectional airflow that may be channeled and eventually converged for modulation. Initial modulation of the air flow may be performed via constricting valves with multiple phased chambers (described below with respect to FIGS. **8-9**). Additionally or as an alternative, the flow may be further modulated where the bidirectional flow converges using a micro-valve array as described above. Such modulation(s) may be tuned to produce high quality low frequency sound.

(44) Flow can be controlled via the high frequency source by: (i) an amplitude of the drive; (ii) a frequency of drive; and/or (iii) a ratio of the source/sink pump drive. However, care should be taken to adjust for interference, and sum and difference of the frequency interactions between source and sink. Furthermore, as is similarly known with a bipolar junction transistor (BJT) amplifier, care must be taken when switching from source to sink and vice-versa. Accordingly, direction of air flow may be bidirectional depending on which of the pumps is operating or which is being driven harder.

(45) FIGS. **5A** and **5B** depict examples of sheet arrays included in a fluid pump to produce an air flow. As opposed to the rigid array of FIG. **4**, the sheet array **500** has a translated lag at the floating end **502**. Referring to FIG. **5A**, a timed series (1-4) shows the arrays of sheets, all of which are anchored at one edge **504**, over the course of a drive cycle. As shown, an anchored edge **504** is driven by the high frequency source, causing the floating edge **502** to produce air flow, and creating a similar air-grate-pump effect. The amount of bend or translated lag is dependent on material, thickness of each sheet, the length of edges, and a frequency/amplitude of drive signal. Such implementations can be very lightweight. Furthermore, using similar opposing grate-pumps, bidirectional flows can also be created.

(46) FIGS. **6A** and **6B** illustrate a bidirectional arrangement for a sheet array included in a fluid pump to produce a bidirectional air flow. The arrangement as shown in FIG. **6A** includes a source pump **600** and a sink pump **602**, each including an oscillatory drive source **604**, **606** to drive the arrays in order to produce the bidirectional air flow. For example, drive source **604** is shown as the driving mechanism for the source pump **600**, and drive source **606** is the driving mechanism for the sink pump **602**. As such, the source pump **600** and the sink pump **602** comprise a bidirectional array including a source sheet array **608** having an anchored end **612** and a floating end **616**, and a sink sheet array **610** having an anchored end **614** and a floating end **618**. This bidirectional “waffle” arrangement allows for flow in either direction depending on the driving mechanisms, whereby the sink sheet array **610** is orthogonal to the source sheet array **608**, and where the anchored ends **612**, **614** of the source sheet array **608** and the sink sheet array **610** are tied to each other. By combining both amplitude and frequency modulation of either or both grids of the waffle arrangement the system can produce sounds at any desired audio frequency even with the drive frequencies are in the high ultrasonic range. Although not illustrated to maintain visual clarity, small supporting structures can be attached to the waffle structure to facilitate better conduction from actuators to the waffle blades.

(47) FIG. 6B illustrates perspective views of the bidirectional “waffle” arrangement of FIG. 6A. The “left side view” **620** shows the source pump array **624** anchored to a first sheet **622** of the sink array **610**. The drive source **604** for the source pump **600** can oscillate the source pump array **624** at a high frequency to produce a steady airflow in an upward direction. Each sheet in the source array **624** may be tied to a corresponding sheet in the sink array **610**. Similarly, the “front side view” **626** shows the sink pump array **628** anchored to a first sheet **630** of the source pump array **624**. Also, each sheet in the sink pump array **628** may be tied to a corresponding sheet in the source array **608**. Thus, air flow may be produced by driving the sink pump **602** at a high frequency to produce a steady air flow in a downward direction. As discussed above, such flows may be produced simultaneously and converged. Alternatively, periodically alternating flows may be generated and modulated at a single convergence zone in a sound channel.

(48) FIG. 7A and FIG. 7B illustrate a wave grid showing a rotational single directional version of the waffle blade topology. FIG. 7B depicts a three-dimensional illustration of the blade in relief for visual clarity with support and actuator structures omitted. Actuators on either side of the fanned out blades cause the blades to move in rotational manner both clockwise and/or counter clockwise about a middle point in center of the actuators. Audio frequencies are achieved via frequency modulation (FM) and amplitude modulation (AM) in combination via the controller in FIG. 1. Supporting structure is shown in FIG. 7A and FIG. 7B to help conduct force from the actuators through the blade assembly.

(49) FIGS. 8A and 8B illustrate chamber-based modulators **800**, **810** (having four chambers and three chambers respectively) for producing phase-controlled air flow. Referring to FIG. 8A, the chamber-based modulator **800** includes four chambers, and is capable of producing a variety of drive phases. Each chamber in the chamber-based modulator **800** can be driven via an electric field, a magnetic field, mechanically, piezo-electrically, or by way of any combination listed. As shown, the chambers of the modulator **800** can be manipulated (as shown in phases  $\phi 1$ - $\phi 4$ ) to modulate airflow being produced by the fluid pump. Such modulation may be performed prior to the flow interacting with the micro-valve array. Alternatively, the micro-valve array itself may be composed entirely of such modulators **800**, which may be actuated in conjunction by way of the electric or magnetic field, mechanically, piezo-electrically, or otherwise.

(50) Similarly, FIG. 8B shows a chamber-based modulator **810** that includes three chambers (six drive phases are shown,  $\phi 1$ - $\phi 6$ ). The chamber-based modulator **810** can also be driven via an electric field, a magnetic field, mechanically, piezo-electrically, or by way of any combination listed. Similarly still, the modulator **810** is shown as a “constrictor” valve, and can be included with the fluid pump for pre-sound modulation, and/or may comprise the entire micro-valve array as discussed above. Implementations requiring post-modulation to produce audible frequencies can include the micro-valve array, which can be synchronized with the chamber-based modulator **810** in order to produce more precise or refined pressure pulses.

(51) FIG. 9 illustrates an implementation of a chambered modulator **900** for producing a phase-controlled air flow. Four phases are shown, and include various stages of constriction and release in order to provide a modulated air flow. However, many more phases are possible with a four chambered modulator **900** to produce varying frequency pressure pulses. Phase one ( $\phi 1$ ) shows fluid from a fluid pump entering the modulator through its input side **902** and compressing against the third chamber. A pulse is formed by closing off the modulator and driving the compressed fluid through the modulator as shown in phases two and three ( $\phi 2$ ,  $\phi 3$ ). As shown in phase four ( $\phi 4$ ), the compressed fluid is driven out of output side **904** the modulator **900**. For AC implementations, a post modulating signal **906** may be applied in addition to the chambered pulse in order to further refine control of the air flow.

(52) FIG. 10 illustrates an example electrical schematic for a four-chambered modulator. As shown in FIG. 10, a pair of transformers can be configured to trigger the operation of each chamber in the four-chambered modulator as shown in FIG. 9. The circuit **1000** may trigger each chamber

according to an applied electric or magnetic field. Additionally or as an alternative, the modulators may be at least partially comprised of piezo-electric material, and may react to pressure, mechanical, and/or electromagnetic signals. Each chamber in the modulator may be triggered accordingly to a different signal, or may be synchronized to another chamber to operate in unison with one or more other chambers in the modulator.

(53) FIG. 11 illustrates the pump apparatus 1100 mounted inside a housing 1110 with a small chamber 1120 and exit holes 1130. The chamber and exit holes may be tuned to provide a low-pass filter in conjunction with standard acoustic practice to improve frequency response and uniformity for audio quality purposes. For example, each of the exit holes may be tuned for a specified audible frequency.

## CONCLUSION

(54) It is contemplated for embodiments described herein to extend to individual elements and concepts described herein, independently of other concepts, ideas or system, as well as for embodiments to include combinations of elements recited anywhere in this application. Although embodiments are described in detail herein with reference to the accompanying drawings, it is to be understood that this disclosure is not limited to those precise embodiments. As such, many modifications and variations will be apparent to practitioners skilled in this art. Accordingly, it is intended that the scope of this disclosure be defined by the following claims and their equivalents. Furthermore, it is contemplated that a particular feature described either individually or as part of an embodiment can be combined with other individually described features, or parts of other embodiments, even if the other features and embodiments make no mention of the particular feature. Thus, the absence of describing combinations should not preclude the inventor from claiming rights to such combinations.

(55) One or more embodiments described herein provide that methods, techniques and actions performed by a computing device are performed programmatically, or as a computer-implemented method. Programmatically means through the use of code, or computer-executable instructions. A programmatically performed step may or may not be automatic.

(56) One or more embodiments described herein may be implemented using programmatic modules or components. A programmatic module or component may include a program, a subroutine, a portion of a program, or a software component or a hardware component capable of performing one or more stated tasks or functions. As used herein, a module or component can exist on a hardware component independently of other modules or components. Alternatively, a module or component can be a shared element or process of other modules, programs or machines.

(57) Furthermore, one or more embodiments described herein may be implemented through the use of instructions that are executable by one or more processors. These instructions may be carried on a computer-readable medium. Machines shown or described with FIGS. below provide examples of processing resources and computer-readable mediums on which instructions for implementing embodiments can be carried and/or executed. In particular, the numerous machines shown with embodiments include processor(s) and various forms of memory for holding data and instructions. Examples of computer-readable mediums include permanent memory storage devices, such as hard drives on personal computers or servers. Other examples of computer storage mediums include portable storage units (such as CD or DVD units), flash memory (such as carried on many cell phones and tablets)), and magnetic memory. Computers, terminals, network enabled devices (e.g., mobile devices such as cell phones) are all examples of machines and devices that utilize processors, memory and instructions stored on computer-readable mediums. Additionally, embodiments may be implemented in the form of computer-programs, or a computer usable carrier medium capable of carrying such a program.

(58) Although illustrative embodiments have been described in detail herein with reference to the accompanying drawings, variations to specific embodiments and details are encompassed by this disclosure. It is intended that the scope of the invention is defined by the following claims and their

equivalents. Furthermore, it is contemplated that a particular feature described, either individually or as part of an embodiment, can be combined with other individually described features, or parts of other embodiments. Thus, absence of describing combinations should not preclude the inventor(s) from claiming rights to such combinations.

(59) While certain embodiments have been described above, it will be understood that the embodiments described are by way of example only. Accordingly, this disclosure should not be limited based on the described embodiments. Rather, the scope of the disclosure should only be limited in light of the claims that follow when taken in conjunction with the above description and accompanying drawings.

## Claims

1. A sound generation system comprising: a controller; a fluid pump connected to the controller to receive commands, the fluid pump being operable to produce an air flow through a sound channel; and a sound control medium disposed within the sound channel, wherein the sound control medium receives and modulates the airflow to produce pressure waves at a plurality of audible frequencies, the sound control medium including a valve redirector that is operable to redirect a back flow of the air flow through one or more waste channels in accordance with a desired audible frequency of the plurality of audible frequencies.
  2. The sound generation system of claim 1, wherein the sound control medium includes a grating disposed within the sound channel.
  3. The sound generation system of claim 2, wherein the grating is comprised of a first micro-valve array disposed on a plurality of openings of the grating.
  4. The sound generation system of claim 3, wherein each micro-valve of the first micro-valve array is configured to operate in unison to collectively manipulate the air flow to produce pressure waves at a plurality of audible frequencies.
  5. The sound generation system of claim 1, wherein at least one waste channel of the one or more waste channels includes waste channel grating disposed within the waste channel.
  6. The sound generation system of claim 5, wherein the waste channel grating disposed within the at least one waste channel includes a micro-valve array disposed on a plurality of openings of the waste channel grating.
  7. The sound generation system of claim 1, wherein the controller generates valve control signals that dynamically and synchronously control both the sound channel and waste channel.
  8. The sound generation system of claim 1, wherein the controller processes audio signals and implements commands to the fluid pump and the sound control medium.
  9. The sound generation system of claim 1, wherein the waste channel includes a second micro-valve array.
  10. The sound generation system of claim 1, wherein the fluid pump includes a single array of air gates, wherein the single array is driven by a high frequency source which can generate airflow that can be both amplitude and frequency modulated to produce pressure pulses.
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